# Mapping the Machine You Live On

COL334 Computer Networks — Assignment 1, Part A

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# Abstract

This report covers Part A of COL334 Assignment 1, in which we measure and document the network infrastructure of the IIT Delhi campus from the vantage point of our own devices. Working as cartographers, we traced routes to a fixed set of targets and to a further set of campus and external destinations, isolated the internal (`10.x.x.x`) hops that recur across those traces, and used position, recurrence, and round-trip time to infer the likely role of each hop in the campus topology. We surveyed the wireless layer from three locations, recording every visible SSID, BSSID, band, channel, and signal strength, and used this to determine how many physical access points are actually behind the broadcasts and how the campus's channel plan is enforced. We then ran a set of ping-based delay experiments – fixed-size probes to five targets at increasing distance, a packet-size sweep to isolate transmission delay, and (where data was available) a queue-occupancy comparison between busy and quiet hours – to separate propagation, transmission, and queuing delay in practice. Throughout, we have tried to keep the line between what we directly measured and what we are inferring explicit, and to flag alternative explanations we could not rule out rather than presenting a single tidy story.

# Acknowledgements

We thank the COL334 teaching staff for the assignment design and the shared class map, and IIT Delhi's network infrastructure for tolerating two days of low-rate probing.

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# Chapter 0

# Introduction

Assignment 1 for COL334 (Computer Networks) asks every team to map as much of the IIT Delhi campus network as can honestly be measured from an ordinary student device, without authentication attempts, SNMP, vulnerability scanners, aggressive scan timing, or scanning of anyone else's device. This report covers **Part A: Cartography**, worth 50 of the 100 marks for the assignment. Part B (the traffic-capture "stakeout") is being completed and reported separately by our teammate.

## 0.1 Team

This report was prepared jointly by:

- Tejasvi Shrivastava (2024CS10582)
- Yashvardhan Singh Chaudhary (2024CS10300)

## 0.2 Rules of Engagement

All measurements in this report were taken under the constraints set out in the assignment handout: no authentication attempts, no SNMP, no exploits or vulnerability scanners, scan timing no more aggressive than `-T2`, and no scanning of devices that might belong to a student or staff member rather than to the infrastructure itself.

## 0.3 Fixed Targets

Wherever this report refers to "the targets", it means the five fixed destinations specified in the handout:

- **T1** – our default gateway (found per-device, see Chapter 1)
- **T2** – `www.iitd.ac.in`
- **T3** – `8.8.8.8`
- **T4** – `www.iitb.ac.in` (∼1100 km)
- **T5** – `www.mit.edu` (∼12,000 km)

## 0.4 Chapter List

**Chapter 1, Your Path Out.** Our own device-to-Internet path: interface, gateway, DNS, and the first hops out of campus, each tagged with evidence.

**Chapter 2, Campus Topology.** The router hunt (A2.1), the wireless survey (A2.2), and our contribution to the shared class map (A2.3).

**Chapter 3, The Delay Experiment.** Where RTT time goes (A3.1), making transmission delay visible via a packet-size sweep (A3.2), and watching a queue fill (A3.3).

**Chapter 4, Evidence.** The numbered evidence items (E-1 onward) that the rest of the report cites.

**Chapter 5, Findings.** Three specific, checkable findings about the network, each with an alternative explanation we could not rule out.

## 0.5 A Note on Honesty

The assignment explicitly rewards the distinction between what we observed directly and what we are inferring, and rewards an honestly reported "no difference" or "couldn't reach" over a tidied-up positive result. We have tried to keep to that standard throughout – flagging inferred elements, reporting dropped probes as data rather than omitting them, and naming a second explanation wherever our evidence does not fully rule one out.

# Chapter 1

# Your Path Out

This chapter documents the one part of the network we can see completely: the path from our own device out to the open Internet.

## 1.1 Device and Access Point

Measurements in this section were taken from a laptop running Ubuntu, connected over Wi-Fi (interface `wlo1`) from room NB-09, Satpura Hostel. The access point nearest us is photographed in Evidence E-1 (Figure 1.1), with a handwritten placard giving our roll numbers, the date, and the location.

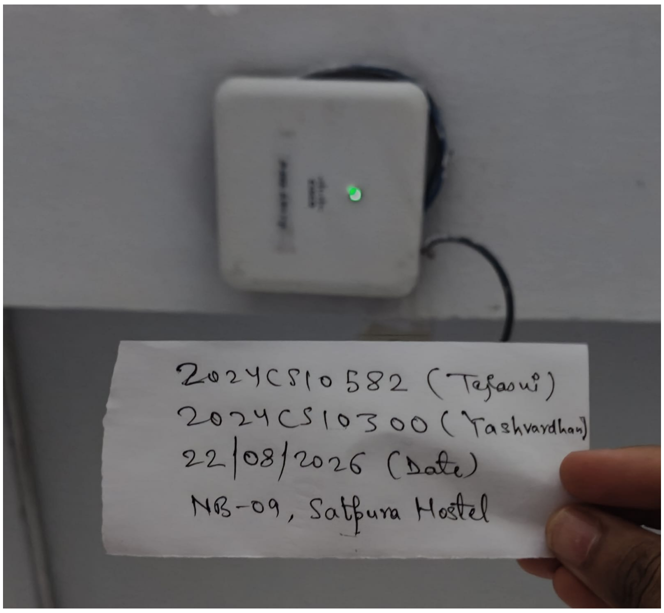


Figure 1.1: The access point/wall unit nearest our device during measurement, with evidence placard. See Evidence E-1.

## 1.2 Finding T1

`ip -br a` shows our device's address as `10.184.6.40/19` on `wlo1`. `ip route | grep default` gives:

```
default via 10.184.0.1 dev wlo1 proto dhcp src 10.184.6.40 metric 600
```

so the DHCP-assigned default gateway is **10.184.0.1**. `resolvectl status` shows our active DNS resolver as **10.7.172.202** (with 10.10.1.4, 10.10.1.5, 10.10.2.2 also configured).

**A discrepancy worth flagging.** The handout notes that a traceroute to anything outside campus should show the gateway at hop 1 "by definition", and that if it doesn't, that's worth a sentence. In our case it doesn't: every traceroute we ran (Chapter 2, and the one below) shows hop 1 as **10.184.0.13**, not 10.184.0.1. Both addresses are in the same `/19`, so our best explanation is that 10.184.0.1 is a virtual/first-hop address (e.g. a shared gateway IP fronting several access switches, possibly via VRRP-style redundancy), while 10.184.0.13 is the specific physical router interface that actually decrements TTL and replies with the ICMP time-exceeded message – i.e. our packets' first real Layer-3 hop is a different, specific box behind the advertised gateway address. We cannot fully rule out the alternative that 10.184.0.1 simply never responds to traceroute probes (rate-limited or filtered ICMP) while still being the true first hop – we have no way to distinguish "different device" from "same device, doesn't reply on this address" without further tooling that is out of scope here. For the rest of this report we follow the handout's implicit intent and use **10.184.0.13** as T1, since that is the address that is consistently and observably our first hop.

## 1.3 Path to the Boundary

A traceroute to an external destination not previously used elsewhere in this report (`www.codechef.com`, chosen to avoid double-counting a target already used for T1–T5) gives:

```
1  10.184.0.13  4.929 ms 4.856 ms 4.816 ms
2  10.255.109.104.761 ms 4.732 ms 4.702 ms
3  10.255.107.3 4.679 ms 4.649 ms 4.622 ms
4  10.119.233.654.590 ms 4.560 ms 4.531 ms
5  * * *        (unresponsive)
6  10.119.234.168.483 ms 7.723 ms 7.581 ms
7  * 14.143.159.145 11.758 ms * (first public hop)
8  * * *        (unresponsive)
9  121.240.255.311.581 ms 11.539 ms *
10 104.23.231.30917836msms * 104.23.231.11
```

```
11 172.67.75.52 9.713 ms 8.882 ms 8.770 ms   (destination)
```

Hops 1–6 are all internal (`10.x.x.x`) and match the core/egress path already identified in Table 2.1 (Chapter 2): local gateway, the two core distribution routers, the internal border router, and the external egress gateway. Hop 5 does not respond, which we read as an unresponsive internal hop rather than a gap in the path, since hop 6 replies with a plausible, monotonically-consistent RTT. The first *public* IP appears at hop 7:

Table 1.1: Whois lookups for the first public hops after our gateway.

| Hop | IP | Owner (whois) | Role |
|---|---|---|---|
| 7 | 14.143.159.145 | Tata Communications (TATACOMM-IN) | Upstream ISP hop |
| 9 | 121.240.255.30 | Tata Communications (TATACOMM-IN, ex-VSNL) | Upstream ISP hop |
| 10 | 104.23.231.30 / .11 | Cloudflare, Inc. (CLOUDFLARENET) | CDN edge |

This is where our traffic crosses the IIT Delhi boundary: hop 6 (10.119.234.162, the internal egress gateway) hands off to hop 7, the first Tata Communications address. From there it stays on Tata's network (hop 9) before landing on a Cloudflare edge node (hop 10) that terminates `www.codechef.com` – consistent with the CDN-edge explanation we gave in Chapter 3 for why T5's measured minimum RTT undercut its nominal geographic distance: Tata Communications appears to be (at least one of) our commercial upstream(s), and a lot of what looks like "the open Internet" from here is actually a nearby CDN edge rather than the origin server.

## 1.4 Diagram: Device to Internet

TODO: one diagram (Graphviz, style of Figure 2.1), built from the data above: device → wlo1 → AP (E-1) → 10.184.0.13 (T1) → 10.255.109.100 → 10.255.107.3 → 10.119.233.65 → [10.184.0.1 aside, see discrepancy note] → 10.119.234.162 → **boundary** → 14.143.159.145 (Tata) → 121.240.255.30 (Tata) → 104.23.231.x (Cloudflare), with a dashed "here my evidence stops" line after the third public hop and everything past it in grey/*inferred*. Both teammates' paths side by side – need Yashvardhan's equivalent data to add his side.

## 1.5 Evidence Tags

Every element in the final diagram must carry an evidence tag (E-1, E-2, . . . ) referencing Chapter 4; untagged elements score zero per the assignment's marking rules.

# Chapter 2

# Campus Topology

## 2.1 A2.1 – The Router Hunt

Traceroutes were run to at least 10–15 destinations: campus services (`www.iitd.ac.in`, `moodle`, `library`, `webmail`, `cse`, `am`, `chemical`, `design`, `dms`, `textile`, `ee`, `maths`) and external destinations (`1.1.1.1`, `8.8.8.8`, `www.iitb.ac.in`, `www.mit.edu`, and several large commercial sites used purely to sample more of the egress path). Table 2.1 lists every distinct internal (`10.x.x.x`) hop observed, an abbreviated view of which traces it appeared in, its position along the path, its average RTT, and our best guess at its role, reasoned from that position, recurrence, and RTT together.

Table 2.1: Observed internal hops and inferred roles. "Traces" is abbreviated to a representative subset with a count; the full per-trace list for each hop is preserved as Evidence E-2.

| Hop IP | Representative traces (count) | Pos. | Avg RTT (ms) | Best guess at role |
|---|---|---|---|---|
| 10.184.0.13 | www.iitd.ac.in, moodle, cse, 8.8.8.8, www.iitb.ac.in, ... (24) | 1 | 4.45 | Local gateway (T1) for the hostel area |
| 10.254.236.26 | am, webmail, textile, dms, moodle, www.iitd.ac.in, design (7) | 2 | 5.09 | Primary aggregation router (campus services) |
| 10.254.236.6 | ee, library, chemical, maths (4) | 2 | 4.55 | Secondary aggregation router |
| 10.10.17.1 | moodle (1) | 3 | 3.62 | Department / service subnet gateway |
| 10.254.208.6 | cse (1) | 2 | 7.17 | Department subnet gateway (CSE) |
| 10.208.20.2 | cse (1) | 3 | 7.17 | Department subnet gateway (CSE, internal) |
| 10.10.211.213 | am (1) | 3 | 8.44 | Department / service subnet gateway |
| 10.7.10.101 | chemical (1) | 3 | 6.22 | Department / service subnet gateway |
| 10.10.17.2 | design (1) | 4 | 6.13 | Department / service subnet gateway |
| 10.10.211.211 | textile (1) | 3 | 7.51 | Department / service subnet gateway |
| 10.10.211.216 | maths (1) | 3 | 4.96 | Department / service subnet gateway |
| 10.255.109.100 | whatsapp, amazon, instagram, google, netflix, mit.edu, ... (15) | 2 | 5.01 | Campus core distribution router |
| 10.255.107.3 | (as above) + www.iitb.ac.in (16) | 3 | 5.39 | Campus core distribution router |
| 10.119.233.65 | (as above) (16) | 4 | 5.99 | Internal border / transit router |
| 10.119.234.162 | amazon, ibm, netflix, mit.edu, wikipedia, x, microsoft, reddit, apple (9) | 6 | 8.55 | External egress gateway (commercial ISP path) |
| 10.1.207.69 | whatsapp, instagram, iitb.ac.in, google, facebook, 8.8.8.8 (6) | 5 | 29.22 | Transit router (NKN / peering path) |
| 10.1.200.137 | whatsapp, instagram, iitb.ac.in, google, facebook, 8.8.8.8 (6) | 6 | 34.58 | Transit router (NKN / peering path) |

Note: position/RTT reconstructed from a flattened source table – verify against the original CSV before final submission (likely viva material).

A few observations follow directly from the table. `10.184.0.13` appears at position 1 in every trace we ran – consistent with it being our own default gateway rather than a shared campus device. The pair `10.255.109.100` / `10.255.107.3` / `10.119.233.65` appears in every trace to an external commercial destination but never in traces to `www.iitd.ac.in` or other campus services, which is the signature of a core/egress path that campus-internal traffic never needs to cross. Two hops, `10.1.207.69` and `10.1.200.137`, sit at a noticeably higher average RTT (29–35 ms) than the rest of the internal hops (4–9 ms); combined with their appearing only on traces toward external destinations and not toward `www.iitb.ac.in` exclusively, we read them as the peering/transit hops just before or on IIT Delhi's connection to the National Knowledge Network (NKN) or a commercial upstream, rather than anything internal to a single building. The department gateways (`10.10.17.x`, `10.10.211.x`, `10.208.20.2`, `10.7.10.101`, `10.254.208.6`) each appeared in exactly one trace, which is expected: each department's traceroute only crosses its own subnet gateway.

### 2.1.1 Internal Topology Graph

Figure 2.1 draws the internal topology as a graph, with routers as nodes and observed adjacencies (which hop directly followed which, across all traces) as edges.

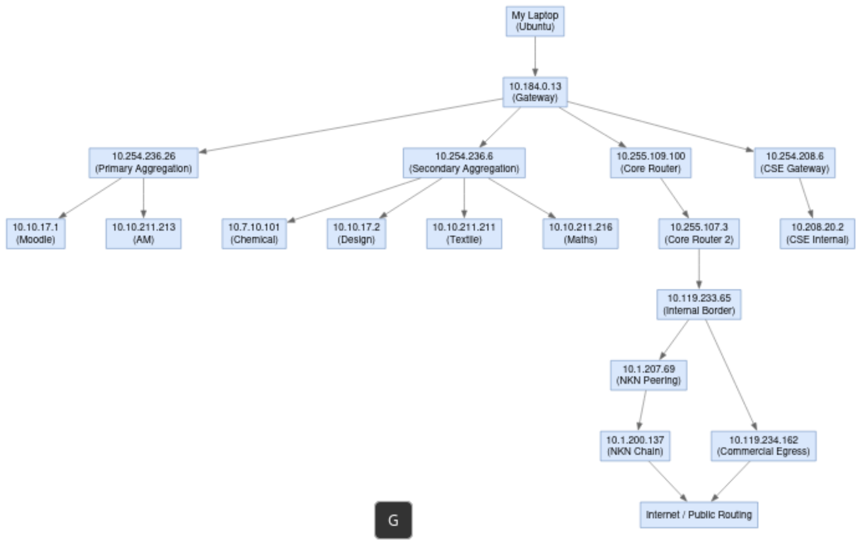


Figure 2.1: Internal topology graph built from observed adjacencies in Table 2.1. Edges represent a hop directly following another in at least one trace; nothing past the last labelled node was directly observed.

## 2.2 A2.2 – The Wireless Layer

A Wi-Fi survey was run from three locations on campus using `nmcli`, recording every broadcast SSID together with its BSSID, band, channel, and signal strength.

Table 2.2: Wireless survey across three locations (redundant enterprise BSSIDs at Satpura truncated for length; full capture preserved as Evidence E-3).

| Location | BSSID | SSID | Band | Ch. | Sig. |
|---|---|---|---|---|---|
| Satpura | 4E:BC:FE:A5:A4:A4 | realme P2 Pro 5G | 2.4 GHz | 6 | 97 |
| Satpura | E4:4E:2D:0C:82:61 | eduroam | 2.4 GHz | 1 | 74 |
| Satpura | E4:4E:2D:0C:82:60 | IITD_WIFI | 2.4 GHz | 1 | 74 |
| Satpura | E4:4E:2D:0C:82:64 | IITD_Secure_GUEST | 2.4 GHz | 1 | 74 |
| Satpura | E4:4E:2D:0C:82:63 | (Hidden) | 2.4 GHz | 1 | 74 |
| Satpura | E4:4E:2D:0C:82:6E | eduroam | 5 GHz | 44 | 70 |
| Satpura | E4:4E:2D:0C:82:6F | IITD_WIFI | 5 GHz | 44 | 69 |
| Satpura | E4:4E:2D:0C:8B:E1 | eduroam | 2.4 GHz | 6 | 59 |
| Satpura | A4:B4:39:40:C3:61 | eduroam | 2.4 GHz | 11 | 59 |
| Satpura | A4:B4:39:33:7E:A4 | IITD_Secure_GUEST | 2.4 GHz | 11 | 40 |
| Satpura | 88:9C:AD:5A:43:41 | eduroam | 2.4 GHz | 11 | 29 |
| Lecture Hall | 54:8A:BA:DD:7A:C1 | IITD_Secure_GUEST | 2.4 GHz | 6 | 97 |
| Lecture Hall | 54:8A:BA:DD:7A:C4 | IITD_WIFI | 2.4 GHz | 6 | 87 |
| Lecture Hall | 54:8A:BA:DD:7A:C3 | IITD_IOT | 2.4 GHz | 6 | 87 |
| Lecture Hall | 54:8A:BA:DD:7A:CE | IITD_Secure_GUEST | 5 GHz | 36 | 82 |
| Lecture Hall | 04:5F:B9:2F:24:C1 | IITD_Secure_GUEST | 2.4 GHz | 1 | 69 |
| Lecture Hall | 70:BD:96:6E:75:42 | eduroam | 2.4 GHz | 11 | 65 |
| Lecture Hall | 38:7A:CC:6D:56:A9 | 3E4C54F42B90 | 5 GHz | 44 | 59 |
| Lecture Hall | 78:22:88:40:9F:42 | 454B10749644 | 5 GHz | 44 | 44 |
| Lecture Hall | 04:5F:B9:2F:24:CB | IITD_WIFI | 5 GHz | 116 | 42 |
| Library | 70:DF:2F:04:98:62 | eduroam | 2.4 GHz | 1 | 72 |
| Library | 70:DF:2F:04:98:63 | IITD_IOT | 2.4 GHz | 1 | 72 |
| Library | 5C:A6:2D:44:E7:E2 | eduroam | 2.4 GHz | 11 | 62 |
| Library | 5C:A6:2D:44:E7:ED | eduroam | 5 GHz | 128 | 56 |
| Library | 70:DF:2F:04:98:6C | IITD_IOT | 5 GHz | 149 | 52 |
| Library | B8:B7:DB:E6:BD:B5 | 26582626 | 2.4 GHz | 5 | 42 |
| Library | BE:00:19:A0:0C:FA | rjnish | 2.4 GHz | 6 | 40 |
| Library | 5C:A6:2D:6F:F8:E1 | IITD_Secure_GUEST | 2.4 GHz | 1 | 39 |

### 2.2.1 Reading the Survey

**How many physical APs.** SSIDs are not access points – a single radio commonly broadcasts several SSIDs from one MAC block (multiple BSSID / MBSSID), differing only in the last hex digit. Grouping BSSIDs by their shared prefix rather than counting SSID rows gives 5 distinct physical enterprise APs in Satpura, 3 in the Lecture Hall, and 4 in the Library – noticeably fewer than the raw broadcast count.

**Global vs. localised SSIDs.** `eduroam`, `IITD_WIFI`, and `IITD_Secure_GUEST` appear in all three locations and are evidently provisioned campus-wide. `IITD_IOT` appears in both academic locations (Lecture Hall, Library) but not once in the Satpura data, which reads as a deliberate choice to keep IoT infrastructure out of residential buildings rather than a coverage gap – though we cannot rule out that our single Satpura vantage point simply missed an IITD_IOT radio elsewhere in the hostel. Personal hotspots (`realme P2 Pro 5G`, `rjnish`, the two unnamed hex-string SSIDs in the Lecture Hall) are, as expected, seen at exactly one location each and nowhere else.

**Channel plan.** Across all three surveys, zero networks were observed on 2.4 GHz channels 3, 4, or 8 – every enterprise AP sits on 1, 6, or 11, the three mutually non-overlapping channels. This is consistent with the campus enforcing a fixed channel plan (RRM) on managed infrastructure. The one exception is the student hotspot `26582626` in the Library, broadcasting on channel 5, which does create adjacent-channel interference with neighbouring channel-1 and channel-6 networks – but since it is a personal device we cannot scan or otherwise investigate further under the assignment's rules.

## 2.3 A2.3 – The Class Map

TODO: add rows to `COL334_A1_Campus_Map.xlsx`, verify two other teams' rows (confirmed/contradicted/couldn't reach), summarise here.

# Chapter 3

# The Delay Experiment

## 3.1 A3.1 – Where Does the Time Go?

Each of T1–T5 was pinged 50 times; Table 3.1 records the minimum, average, and maximum RTT observed for each. Distances are rough straight-line estimates – T2/T4 from map distance to the named campus, T3/T5 from public IP geolocation (the resolved IP's registered location) – and are not independently verified against ground truth.

Table 3.1: Aggregate RTT statistics, 50 packets per target.

| Target (role) | Resolved IP | Est. distance | Min / Avg / Max RTT (ms) |
|---|---|---|---|
| T1 (local gateway) | 10.184.0.13 | $<$0.1 km | 3.904 / 8.891 / 96.290 |
| T2 (campus target) | www.iitd.ac.in | $\sim$2.0 km | 3.315 / 7.489 / 14.110 |
| T3 (regional DNS) | 8.8.8.8 | $\sim$20.0 km | 42.824 / 49.859 / 119.830 |
| T4 (IIT Bombay) | www.iitb.ac.in | $\sim$1100 km | N/A (100% loss) |
| T5 (global target) | www.mit.edu | nominal 12,000 km | 35.596 / 51.145 / 247.982 |

T4 returned 100% packet loss across all 50 probes. We read this as ICMP being firewalled at or near `www.iitb.ac.in` rather than a routing failure – traceroutes toward the same destination (Chapter 2) do get responses from intermediate hops, which would not happen if the path itself were broken.

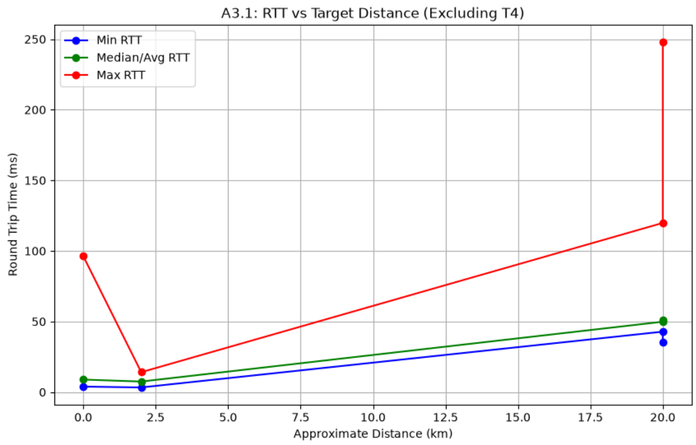


Figure 3.1: Minimum, median/average, and maximum RTT against approximate distance for T1, T2, T3, T5 (T4 excluded – no response).

**Propagation delay vs. measured minimum.** At $2 \times 10^8$ m/s, T2's $\sim$2 km distance implies a propagation delay of about 0.02 ms – negligible next to the measured 3.3 ms minimum, so for a target this close essentially all of the RTT floor is transmission and processing delay, not propagation. T5's 12,000 km distance implies a propagation floor of about 120 ms round-trip, yet the measured minimum was 35.6 ms – well under that floor. The straightforward reading is that our traffic did not actually travel the full nominal distance: `www.mit.edu` is very likely served to us from a CDN edge node much closer than Massachusetts. An alternative we cannot fully rule out is that the geolocation-based distance estimate itself is simply wrong (IP geolocation for large anycast/CDN-fronted services is frequently inaccurate), in which case the "true" distance to whatever server actually answered us is just smaller than 12,000 km rather than the packet having taken an implausible shortcut. Either way, the number confirms we are not talking to a server in Cambridge, MA.

**Min-to-max/average gap is queuing delay.** Propagation and transmission delay for a given path and packet size are effectively fixed, so the spread between the minimum and the average/maximum RTT for the same target is attributable to queuing delay – time spent sitting in router buffers behind other traffic. This is why it shows up as *variation* rather than a constant offset: queue occupancy depends on how much cross-traffic happens to be competing for the same link at the moment our packet arrives, which changes probe to probe. T3 shows the widest such gap (42.8 ms min vs. 119.8 ms max), consistent with a longer, shared path through more congested infrastructure than the one-hop path to T1.

**T1's non-zero RTT.** T1 is one hop away, yet its minimum RTT is 3.9 ms, not 0.

Over Wi-Fi, this floor is built from MAC-layer overhead we do not see in the ICMP timestamp: CSMA/CA clear-channel assessment before the medium can be used, the mandatory DIFS/backoff interval, and (in both directions) the transmission delay of putting the frame on and off the physical layer, plus whatever small forwarding/processing delay the gateway itself adds. None of this is propagation delay – at <0.1 km, propagation is on the order of nanoseconds.

## 3.2 A3.2 – Making Transmission Delay Visible

T1 and T3 were pinged at packet sizes 64, 256, 512, 1024, and 1400 bytes (30 packets each); the minimum RTT at each size was plotted against packet size and fit with a straight line for each target (Figure 3.2).

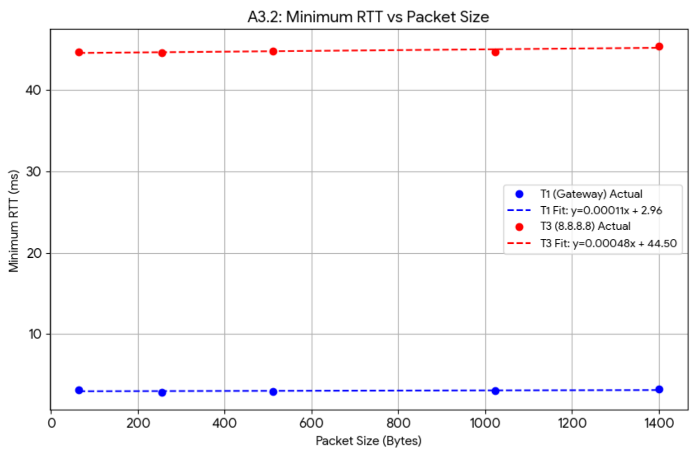


Figure 3.2: Minimum RTT against packet size for T1 and T3, with linear fits.

Fitted lines: T1, $y = 0.00011x + 2.96$; T3, $y = 0.00048x + 44.50$ (RTT in ms, $x$ in bytes).

**The intercept.** The y-intercept is the extrapolated RTT for a 0-byte payload – i.e. everything that is *not* transmission delay for the payload itself: propagation delay, router/host processing, and (for T1) Wi-Fi contention overhead. T3's much larger intercept (44.5 ms vs. 2.96 ms) reflects the extra propagation and per-hop processing accumulated over the longer, multi-hop path to 8.8.8.8, versus the single Wi-Fi hop to T1.

**The slope.** The slope is transmission delay per byte, $L/R$, and since a multi-hop path uses store-and-forward switching, the effective slope is the *sum* of $1/R_i$ over every link on the path, not just the first one. T1's slope (0.00011 ms/byte) is close to flat because one Wi-Fi hop's link rate is high enough that transmission delay for a payload in this size range barely moves the RTT. T3's slope (0.00048 ms/byte) is roughly four times steeper, which is consistent with the packet crossing several additional links (campus core, egress, upstream) each contributing their own $L/R_i$ term, rather than transmission delay being determined by a single bottleneck link. We note that both slopes are small enough, and the underlying ping resolution coarse enough, that we would not want to over-read the precise ratio between them – the qualitative story (T3 accumulates more transmission delay across more hops) is what the data supports; the exact "4×" figure is more fragile.

## 3.3 A3.3 – Watching a Queue Fill

TODO: quiet-hour ping to T3 in hand; busy-hour pending. Once both exist: plot RTT vs. time for both, state which delay component moved (queuing) vs. stayed fixed (propagation/transmission), and whether the queue looks steadily full or spiky. A "no difference" result is valid if explained (e.g. bottleneck may not be campus-internal).

# Chapter 4

# Evidence

The assignment requires 10–15 numbered evidence items, E-1 onward with no gaps, each stating who took it, when, where, the artefact itself, and which map element/table row/plot it supports. What follows is seeded with the items already in hand; the remainder is still to be assembled.

**E-1 – photo**

Who: Tejasvi Shrivastava (2024CS10582), Yashvardhan Singh Chaudhary (2024CS10300)

When: 22-08-2026 [TODO: exact time] Where: NB-09, Satpura Hostel

Artefact: Figure 1.1 – photo of the access point nearest our device, with handwritten placard (roll numbers, date, location).

Supports: Chapter 1, Section "Device and Access Point".

**E-2 – command output / lookup**

Who: [TODO] When: [TODO] Where: [TODO]

Artefact: raw traceroute output / spreadsheet underlying Table 2.1 (full per-trace hop lists, before abbreviation).

Supports: Chapter 2, Table 2.1 and Figure 2.1.

**E-3 – survey**

Who: [TODO] When: [TODO] Where: Satpura Hostel, Lecture Hall, Library

Artefact: full `nmcli` Wi-Fi scan output for all three locations (untruncated).

Supports: Chapter 2, Section A2.2 wireless survey tables.

TODO – remaining items (E-4 through E-10+, up to E-15): raw `ping -c 50` output for T1–T5; packet-size sweep output for T1/T3; quiet/busy-hour continuous-ping logs; `whois` lookups for the first three public hops (Ch. 1); class-map verification screenshots (A2.3). Command output must be verbatim, prompt and errors included.

# Chapter 5

# Three Findings

TODO – exactly three findings, Saw/Evidence/Think/Fix/Check form, at least one from delay (Ch. 3), at least one from topology (Ch. 2). The graded part is the “other explanation” line in Think.

*Candidate 1 (topology):* peering hops 10.1.207.69/10.1.200.137 at 29–35 ms vs. <9 ms for other internal hops.

*Candidate 2 (delay):* T4 dropped 100% of ICMP probes – policy vs. path question.

Format template for each:

> **F-$n$ · ¡title¿**
> Saw: what was measured – a number, not an impression.
> Evidence: E-$x$, E-$y$
> Think: what we conclude – plus one other explanation we couldn’t rule out.
> Fix: specific enough to hand to someone.
> Check: the measurement and threshold that would prove it worked.